

In Class Test - 02

Student Number – 35040

Q1

```
1 public class Vehicle {
2     private int vehicleID;
3     private String brand;
4     private double speed;
5     private boolean isElectric;
6
7     Vehicle(int vehicleID, String brand, double speed, boolean isElectric) {
8         this.vehicleID = vehicleID;
9         this.brand = brand;
10        this.speed = speed;
11        this.isElectric = isElectric;
12    }
13
14    public int getVehicleID() {
15        return vehicleID;
16    }
17
18    public void setVehicleID(int vehicleID) {
19        this.vehicleID = vehicleID;
20    }
21
22    public String getBrand() {
23        return brand;
24    }
25
26    public void setBrand(String brand) {
27        this.brand = brand;
28    }
29
30    public double getSpeed() {
31        return speed;
32    }
33
34    public void setSpeed(double speed) {
35        this.speed = speed;
36    }
37
38    public boolean getIsElectric() {
39        return isElectric;
40    }
41
42    public void setIsElectric(boolean isElectric) {
43        this.isElectric = isElectric;
44    }
45
46    public void displayInfo() {
47        System.out.println("Vehicle ID is: " + vehicleID);
48        System.out.println("Vehicle Brand is: " + brand);
49        System.out.println("Vehicle Speed is: " + speed + "km/h");
50        System.out.println("Electric Power: " + isElectric);
51    }
52 }
53
```



```
1 public class Car extends Vehicle {
2     private int numDoors;
3
4     public Car(int vehicleID, String brand, double speed, boolean isElectric, int numDoors) {
5         super(vehicleID, brand, speed, isElectric);
6         this.numDoors = numDoors;
7     }
8
9     @Override
10    public void displayInfo() {
11        super.displayInfo();
12        System.out.println("Car Doors have: " + numDoors);
13    }
14 }
15
```



```
1 public class Main {
2     public static void main(String [] args) {
3         Vehicle v1 = new Vehicle(101, "BMW", 120.0, false);
4         v1.displayInfo();
5
6         System.out.println();
7
8         Car c1 = new Car(102, "Benz", 240, true, 2);
9         c1.displayInfo();
10    }
11 }
12
```

Q2

```
1 public class Person {
2     private int personID;
3     private String name;
4     private int age;
5     private String email;
6
7     public Person(int personID, String name, int age, String email) {
8         this.personID = personID;
9         this.name = name;
10        this.age = age;
11        this.email = email;
12    }
13
14    public int getPersonID() {
15        return personID;
16    }
17
18    public void setPersonID(int personID) {
19        this.personID = personID;
20    }
21
22    public String getName() {
23        return name;
24    }
25
26    public void setName(String name) {
27        this.name = name;
28    }
29
30    public int getAge() {
31        return age;
32    }
33
34    public void setAge(int age) {
35        this.age = age;
36    }
37
38    public String getEmail() {
39        return email;
40    }
41
42    public void setEmail(String email) {
43        this.email = email;
44    }
45
46    public void displayDetails() {
47        System.out.println("Person ID is: " + personID);
48        System.out.println("Person Name is: " + name);
49        System.out.println("Person Age is: " + age);
50        System.out.println("Person Email is: " + email);
51    }
52 }
53
54
```



```
1 public class Student extends Person {
2     private double gpa;
3
4     public Student(int personID, String name, int age, String email, double gpa) {
5         super(personID, name, age, email);
6         this.gpa = gpa;
7     }
8
9     @Override
10    public void displayDetails() {
11        super.displayDetails();
12        System.out.println("Student GPA is: " + gpa);
13    }
14 }
15
```



```
1 public class Main {
2     public static void main(String[] args) {
3         Person p1 = new Person(101, "Navitha", 19, "navithanavindul@gmail.com");
4         p1.displayDetails();
5
6         System.out.println();
7
8         Student s1 = new Student(102, "Akitha", 22, "akithapasandul@gmail.com", 3.1);
9         s1.displayDetails();
10    }
11 }
12
```